

Builder Journal

Why did you choose AI Engineering?

I chose AI Engineering because Artificial Intelligence is transforming the way people work, learn, and solve problems, and I wanted a structured, hands-on path into it rather than just studying the theory. Across these first assignments, from setting up a professional GitHub presence to researching AI agents, building a working chat application, practicing prompt engineering, and mapping out system architecture, I saw first-hand how AI Engineering combines software development, research, and design into one practical discipline. Understanding how AI models actually work and how they are integrated into real applications convinced me this is the field I want to build my career in.

What type of AI products do you want to build?

I want to build AI-powered applications that solve real-world problems: intelligent chatbots, AI assistants, educational platforms, and productivity tools like the AI Workspace application I built this week. I am also interested in creating AI agents that can automate workflows, use tools, plan multi-step tasks, and coordinate with other agents to perform research or assist businesses in making better decisions. In the future, I hope to contribute to AI solutions that improve people's lives and make advanced technology more accessible.

What was your biggest learning this week?

My biggest learning this week came from building the AI Workspace application, a unified chat interface with system prompts, model selection, prompt templates, and conversation history. It taught me how much state management matters in an AI-powered interface: handling multiple chat sessions, rendering markdown responses correctly, and connecting to a real API all had to work together reliably. Researching the evolution of AI agents alongside this helped me understand LLM applications, tool calling, memory, and planning at a conceptual level, while practicing prompt engineering techniques, role prompting, chain-of-thought, few-shot examples, and structured JSON output, showed me in practice how much control good prompt design gives over a model's behavior.

What challenges did you face?

Most of my biggest challenges came from building the AI Workspace application. Keeping conversation history consistent across multiple chat sessions, rendering markdown responses correctly, and handling edge cases like empty prompts, invalid API keys, rate limits, and connection failures gracefully took a lot of testing. Integrating a real AI API also brought unexpected issues, like requests being rejected because of account credit limits rather than the key itself, which took some debugging to properly diagnose. Alongside this, condensing the research on AI agents into a clear, well-organized report was harder than expected, and designing few-shot prompt examples that generalized well, rather than just repeating the pattern shown, took several iterations to get right.

How did you solve them?

I built and tested the AI Workspace application feature by feature, chat interface, templates, history, error handling, rather than trying to get everything working at once. I deliberately tried empty inputs and bad API responses so I could write clear, honest error messages instead of letting the app fail silently, and traced connection errors carefully to tell apart real bugs from account or API issues. For the research report, I outlined each section before writing so the ideas built logically on one another. For prompt engineering, I tested each technique directly in a notebook, comparing outputs run by run until they were consistent. Reading documentation, practicing consistently, and reviewing concepts again whenever I got stuck helped me build real confidence with these tools.

What are your goals for Week 2?

For Week 2, my goal is to strengthen my Python programming skills and become more comfortable using Git and GitHub for version control. I also want to continue improving my prompt engineering techniques, gain more hands-on experience in building AI applications, and begin developing AI agents that can perform practical tasks. Additionally, I plan to complete all assignments on time, practice consistently, and continue exploring modern AI engineering concepts to build a strong foundation for future projects.